

Gender Gaps in College-Major Sorting: Evidence from Admission Cutoffs*

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Abstract

Can admissions processes contribute to gender gaps in major choices? We study applicants to Colombia’s largest public university who initially report intending the same major and score near its admission cutoff. Women who just miss the cutoff are more likely to secure admission to an alternative major in the first admission cycle, while comparable men are more likely to reapply. This enrollment gap emerges from differences in application choices before current cutoffs are determined and published. Around the eventual cutoff, the discontinuities in the number of majors submitted and in listing a safe first choice are larger for women than for men. These responses are concentrated in majors with small year-over-year change in the cutoff, when the cutoff may plausibly be anticipated. Following applicants in the universe of higher-education enrollment data, we find that marginal women ultimately enroll in majors whose recent graduates have lower earnings, showing that missing the admissions cutoff can generate earnings-relevant sorting.

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1 Introduction

The choice of a college major is highly consequential and can impact future earnings to a similar extent as the decision to enroll in college itself (Kirkebøen et al., 2016; Bleemer and Mehta, 2022). A key finding in the college major choice literature is that women and men sort into different fields, contributing to the gender pay gap among highly educated workers (Black et al., 2008; Sloane et al., 2021). Supply-side determinants like the admissions rules, capacity constraints, and academic thresholds that determine which fields students can access could play a role in this sorting. Existing evidence examines admissions procedures (Patnaik, Wiswall, and Zafar, 2020; Arenas and Calsamiglia, 2025), tuition pricing (Stange, 2015), major restrictions (Bleemer and Mehta, 2021), cutoff-based access to selective programs (Landaud and Maurin, 2025), and gendered responses to grades in introductory courses (Avilova and Goldin, 2018).¹ This paper studies gender differences in applicants' responses to narrowly missing the admission cutoff for their reported intended major using a regression discontinuity design.

Our setting is the admission process at Universidad Nacional, Colombia's largest public university. The university is capacity constrained and highly competitive due to a small number of slots available per major, and a much larger pool of applicants competing for the slots. Priority in the allocation mechanism is determined solely by the university's entrance-exam score. Enrollment at U. Nacional depends on applicants' score, first and second choice majors submitted to the allocation mechanism and remaining slots. We observe applicants' reported intended major, their score, the admissions cutoff score for each major, determined and published after all slots have been filled, and the major in which the applicant ultimately enrolls in at any university in Colombia.

Our identification strategy uses a regression discontinuity design leveraging the fact that

¹A large literature emphasizes demand-side forces behind the gender gap in college major choices: beliefs about earnings (e.g., Wiswall and Zafar, 2015), preferences over non-pecuniary attributes (Zafar, 2013; Ahn et al., 2019; Patnaik et al., 2020), ability (Paglin and Rufolo, 1990; Turner and Bowen, 1999), peer composition (Brenøe and Zölitz, 2020; Calkins et al., 2023), competitiveness (Buser et al., 2014; Landaud et al., 2020), and perceived comparative advantage across subjects (Breda and Napp, 2019; Card and Payne, 2021).

scoring just below or just above the intended-major-specific admissions cutoff is as good as random. “Marginal” applicants are those whose score puts them just below rather than just above the cutoff. Applicants who miss an admission cutoff can either apply to another major at the same university, wait and reapply in a later admission cycle (but must use their new score as old scores are not considered), enroll in another university or exit higher education altogether. Importantly, whether the applicant’s score is above or below the admissions cutoff for their intended major depends on which major they chose to report as their intended major at exam registration, that is, before they know their score. Applicants’ reported intended major is not necessarily a firm commitment or binding preference ranking, but rather information the university collects to estimate demand for different majors, not to allocate slots.

We report regression discontinuity estimates using the distance between the individuals’ score and their intended-major admission cutoff as a running variable. Our main estimate of interest is the “RD gender gap,” or the discontinuity in the gender gap at the cutoff controlling for applicants’ intended major, individual and family characteristic covariates self-reported at registration, and the set of majors available to choose from. We report both the immediate admissions outcomes, occurring before the first cycle’s admissions cutoffs are determined and published, and downstream outcomes, measured after the first admissions cycle.

We find that women who miss the admissions cutoff are more likely to enroll in another major at the same university, while marginal men are less likely to enroll in the current admission cycle and more likely to reapply later. After several cycles, both groups reach similar overall enrollment rates at U. Nacional. The RD gender gap in immediate enrollment is 14.2 percentage points (pp), while in retaking it is -13.2 pp. Our results suggest that these differences reflect how applicants navigate the admissions process. The RD gender gap in the number of majors submitted is 0.16 and in submitting a safe first major choice is 7 pp.

Unlike a typical regression discontinuity context, where the cutoff rule is published in advance and stable over time, college major admission cutoffs can vary over time and may

or may not be known or anticipated by applicants. In our setting, applicants can know their exam score and the previous year’s cutoff, but not the current cutoff for each major, which is determined and published after all slots are filled in that admission cycle. Surprisingly, we find differential discontinuities by gender in the number and type of majors submitted and in immediate enrollment at U. Nacional *even before* the current year’s cutoff score is published, consistent with an anticipatory response.

We then report outcomes downstream of the first admissions cycle, by which point all applicants will have learned if they made or missed the first cycle’s cutoff. The discontinuity in the enrollment gender gap at U. Nacional closes over time, falling from 14.2 pp (significant at the 1% level) to a non-significant 2.0 pp after four admission cycles (two years later). Using data on the universe of higher education enrollment in Colombia, we find no significant RD gender gap in the likelihood of enrolling in applicants’ intended major at any university within two years, or that they substitute toward other universities; if anything, the RD gender gap in enrolling at another university is -9.8 pp (significant at the 10% level).

To investigate potential implications of these differential approaches to the admission process, we look at the potential earnings of the majors applicants enroll in at U. Nacional and at other universities within the next 2 years. Using administrative earnings data for recent graduates, provided to us by the Ministry of Education, we assign each institution-major pair a measure of potential earnings for that major. Marginal women enroll in majors with potential earnings 4.5–4.8% lower than those of marginal men, both within Universidad Nacional and at other institutions.² Using the earnings records of recent graduates, these results are meant to be descriptive of the types of majors men and women sort into and reflect field-level differences, not the marginal effect of narrowly missing a cutoff. They suggest that missing the admission cutoff can generate earnings-relevant sorting.³

Finally, to further investigate the puzzling finding that marginal women “anticipate”

²In Colombia, the gender pay gap among individuals with post-secondary education is 11.8% (National Ministry of Education, 2017).

³Data on applicants’ own earnings records is not available. Since within-major earnings correlate with exam performance, this setup likely overstates the difference for the marginal applicant’s own earnings.

missing the intended major cutoff before the cutoff is determined and published, we conduct heterogeneity analysis by intended major for each of the 49 offered majors. The gaps in early enrollment, number of majors submitted, safe first choices, and retaking are concentrated in majors with small year-over-year cutoff changes, i.e., when last year’s cutoff is closer to this year’s cutoff. As the difference between last year’s and this year’s cutoff increases, the gender gaps in the immediate admissions outcomes attenuate. However, the outcomes measured downstream of the first admission cycle do not follow the same pattern, suggesting that, despite gender differences in navigating the admission process, marginal women nonetheless sort into lower-earnings majors.

Our data do not allow us to distinguish underlying mechanisms, for example gender differences in risk aversion or beliefs, or whether women are making different choices under similar conditions, or making rational choices under different ones. If women face higher penalties for delayed entry due to age at graduation, fewer job opportunities, or slower promotion tracks, then enrolling immediately, even in a lower-paying major, could be a rational response. Instead, our findings contribute to the literature by showing that gender gaps in enrolled majors can emerge within the admissions process even among applicants who report intending the same major and even before admission cutoffs are determined and published.

Our study relates to several areas of literature. We build on the descriptive literature documenting gender differences in approaches to college applications. Prior research shows that women, more than men, diversify by including both selective and safe programs ([Delaney and Devereux, 2021](#)), apply to lower-ranked universities ([Saygin, 2016](#)), and, in non-strategy-proof mechanisms, rank less selective colleges first ([Wu and Zhong, 2020](#)). In contrast with these papers, we provide causal evidence that “diversifying” choices by submitting a longer list of majors is not an approach taken by all women or all marginal applicants, but primarily by marginal women.

Our paper is closely related to [Landaud and Maurin \(2025\)](#), who document gender differ-

ences in reapplication to elite science programs in France after missing the admission cutoff. We show that persistence is not the only margin of response.⁴ In a setting with a broad menu of within- and across-university alternatives, applicants can respond to missing a cutoff by retaking, securing a slot in an alternative major in the current cycle, reapplying in a later admissions cycle, applying elsewhere or not applying to any higher-education institution. Recognizing these margins changes the interpretation of gender gaps in reapplication. Lower retaking among women may reflect not only lower persistence, but also substitution toward alternatives that shape sorting across fields. Ultimately, we contribute to the literature on gender gaps in college major choice by showing that institutional features of admissions systems can generate gender differences in important outcomes, including enrolled major choices even among applicants who initially reported intending to pursue the same major.

Our paper is also related to but different from the literature that estimates major specific returns by exploiting major specific admissions cutoffs e.g., (Hastings et al., 2013; Kirkebøen et al., 2016; Heinesen, 2018; Heinesen and Hvid, 2019; Andersen et al., 2020; Daly et al., 2022; Bleemer and Mehta, 2022). While we do not estimate returns to major, this literature highlights the importance of applicants’ next-best alternatives. We show that these alternatives differ by gender around the admission margin and are associated with different potential earnings.

2 Background

We study applicants to the Bogotá campus of Universidad Nacional de Colombia, the country’s most selective public university. Admission is based entirely on performance on the university’s own college entrance exam (CEE) and the applicants’ first and second major choices submitted to the allocation mechanism. There are two admission cycles each year. The students beginning college in February each year belong to the cycle labeled with a 1

⁴Gender gaps in persistence have also been documented in other domains, for example failing to advance in competition (Ellison and Swanson, 2018; Buser and Yuan, 2019) and after losing an election (Wasserman, 2022).

and those beginning in August belong to cycle 2. For the admission cycle we study (2020-1), applicants took the CEE in September 2019 and, if admitted, started college in February 2020. Each program has a fixed number of slots, and all applicants take the same exam. At registration, applicants report an intended major. This reported intended major is not necessarily a firm commitment or a binding preference ranking. It is not used to allocate slots, and applicants may later submit any first and second major choices with remaining slots to the allocation mechanism.

The admissions process has two features that are central to our analysis. First, applicants know their exam score before submitting their first and second major choices, but they do not know the current admission cutoff for any major. A major’s cutoff is determined only after slots are allocated and equals the score of the last admitted applicant. Previous year’s cutoffs are public, however, and applicants can compare their score with cutoffs from earlier admission cycles. Thus, past cutoffs provide a potentially informative, but imperfect, benchmark for applicants.

Second, the timing of the allocation process makes alternative major choices consequential. Applicants with scores above 600 (one standard deviation above the mean) are assigned to one of four priority groups according to their exam score.⁵ Priority groups enter the online platform sequentially. During a 24-hour window, applicants observe which programs still have slots and may submit up to two ranked major choices.⁶ Slots are allocated within each priority group by serial dictatorship. Applicants are ordered by exam score and assigned to their first choice, or to their second choice if the first choice major has no remaining slots. Applicants who are not assigned a slot may submit again with a later priority group, but by then some programs may have filled and empirically few applicants take this option.

Applicants who are not assigned a slot by the mechanism are not enrolled. Importantly, a slot assignment is not an offer of admission, but automatic enrollment to U. Nacional.

⁵The exams is scored with a mean of 500 and a standard deviation of 100. The distribution of standardized scores is in Figure A1.

⁶Appendix A provides additional details on major submissions and slot availability by priority group.

Enrollees are expected to attend the University the following semester. The University bars enrollees who do not attend from applying again in the following two admission cycles and transferring majors once enrolled is almost impossible. The admission cycle ends when all slots are filled. Applicants who are not assigned a slot can decide to reapply in a future admission cycle, in which case they must retake the CEE. When students retake the entrance exam, only their current score counts (they cannot reuse or keep a higher previous score). More details on the higher education system in Colombia and admissions to Universidad Nacional are in Appendix [A](#).

The institutional setting therefore could create a consequential tradeoff for applicants who expect to miss their intended major’s admission cutoff: secure admission to an alternative program in the current cycle, delay admission and retake the exam under a new score, enroll in another university or not seek enrollment in higher education. This tradeoff is especially salient because Universidad Nacional is a highly selective public university, comparable outside options are often private and less financially accessible, and over 90% of applicants in our sample come from low- or middle-socioeconomic backgrounds. Therefore applicants may value admission to U. Nacional even when the available seat is not in the applicant’s reported intended major.

3 Data Sources and Sample

We use several sources of data. We first use administrative records from Universidad Nacional de Colombia of the universe of applicants to the Bogotá campus for the 2020-1 admission cycle. This dataset includes information from every applicant’s registration form and admissions process. Most importantly, the registration data includes applicants’ reported intended major, a variable rarely available in datasets on school or major choice. With this variable, we do not have to infer applicants’ intentions from their previous choices (for example, from the courses they took in high school) and can identify applicants who just made or just missed

the cutoff for their reported intended major. The CEE registration form also asks applicants to report baseline characteristics about themselves including their gender, their schooling and their families. This dataset also includes information about their CEE exam and admission process including their exam score, their priority group, their first and second major choices submitted to the slot allocation mechanism, and which slot they were enrolled in, if any. We next match applicants in the first admission cycle (2020-1) to reapplication records from the three subsequent cycles (2020-2, 2021-1 and 2021-2) to look at reapplications. We call this matched dataset the “Admissions Data.”

Next, we link the Admissions Data with publicly available information on the current year (2020) and previous years’ major-specific cutoffs and number of slots for every major.⁷ Tables A1 and A2 show the cutoffs and number of slots for every major in the first admission cycle (2020-1) and the cycle one year earlier (2019-1). This table shows wide variation across majors in how much the cutoff changes year over year. Out of 49 majors, 15 majors have an absolute difference of less than 5 standardized points, the average change from 2019 to 2020 is 10.10, median 9.45, min 0.18 and max 29.72.

We then link the Admissions Data with data from the National Higher Education Information System (abbreviated SNIES in Spanish) available from 2019-2 to 2021-2 using each applicant’s national ID number. The “SNIES Data” contains individuals’ attendance histories at all higher education institutions in Colombia. For every attendee, every semester, the SNIES data provides information on the attendee’s major name, institution name and institution type (academic or technical). The SNIES allows us to follow individuals who took the Universidad Nacional CEE but did not enroll there.⁸ Within Universidad Nacional, we observe full application outcomes, while for other universities we only see enrollment through SNIES. Because university admissions in Colombia are not centralized, we do not observe applications to other universities, only enrollment. Therefore, the extensive margin is best

⁷Available here: <https://admisiones.unal.edu.co/servicios-en-linea/estadisticas-del-proceso-de-admision/>

⁸While the slot allocation mechanism used by Universidad Nacional enrolls applicants to available majors, some applicants who secure a slot at Universidad Nacional may nonetheless decide to attend another university.

observed within the single-institution setting.

Finally, we use aggregated administrative payroll data provided to us from the Ministry of Education Labor Market Observatory for Education data or the “LMOE Data.”⁹ The payroll data comes from workers who are in the formal sector (i.e., make public pension contributions). For each higher education institution-major pair, the Ministry provided us the number of 2018 graduates whose 2019 annual earnings fell within ranges. We then calculated the average earnings of recent graduates within each major-institution bin using the midpoints of the range and the number of graduates in each range. We then used the average earnings for each major at each institution as a proxy for the potential earnings of that major at a particular higher education institution.

We restrict the sample to applicants to the Bogotá Campus. The Bogotá campus is the largest and most selective of the University’s branch campuses. High demand and capacity constraints raise the stakes for applicants taking the exam. Approximately 61,500 applicants took the University’s entrance exam in the 2020-1 cycle. 41,613 (about 68%) of those applied to the Bogotá Campus, which had only 3,086 slots available.

4 Econometric Strategy

In the main analyses we use a regression discontinuity design to estimate the effects of just missing one’s intended major admission cutoff. The running variable is the applicant’s standardized CEE score centered at the realized cutoff for their intended major (i.e., the major they stated on the registration form *before* taking the CEE) in the 2020-1 admission cycle. By definition, applicants scoring above could have been enrolled in their intended major if they submitted that major to the allocation mechanism, while applicants scoring below could not have, all else equal. However, unlike in a typical regression discontinuity context, the realized 2020-1 cutoff is not determined in advance and is published a few weeks after the

⁹This data comes from the Labor Market Observatory for Education <https://ole.mineduacion.gov.co/portal/>.

admissions cycle ends.

We estimate the following specification:

$$Y_i = \alpha_0 + \beta_0(\text{Below}_i \times \text{Female}_i) + \beta_1(\text{Below}_i \times \text{Score}_i) + \beta_2(\text{Female}_i \times \text{Score}_i) + \beta_3(\text{Below}_i \times \text{Female}_i \times \text{Score}_i) + \beta_4\text{Below}_i + \beta_5\text{Female}_i + \beta_6\text{Score}_i + X_i'\omega + \gamma_{j(i)} + \delta_{p(i)} + \varepsilon_i. \quad (1)$$

Y_i indicates the outcome of interest for applicant i . Below_i is an indicator for the applicant's CEE score being below the realized cutoff of their reported intended major. Female_i is an indicator for being a female applicant. Score_i is the applicant's CEE score centered at the realized cutoff of their reported intended major. The fully interacted model allows the slope of the running variable to differ on either side of the cutoff for men and women. We also include baseline covariates X_i , intended-major fixed effects $\gamma_{j(i)}$, where $j(i)$ denotes the intended major reported by applicant i , and priority-group fixed effects $\delta_{p(i)}$, where $p(i)$ denotes the priority group to which applicant i is assigned and which determines the applicant's choice set.¹⁰

The two coefficients of interest are β_4 on Below_i and β_0 on $\text{Below}_i \times \text{Female}_i$. Because men are the omitted gender category, β_4 captures the discontinuous change in the outcome at the intended-major admission cutoff for men. The coefficient β_0 captures the difference between the discontinuity for women and the discontinuity for men. Therefore, the discontinuity at the admission cutoff is β_4 for men and $\beta_4 + \beta_0$ for women. We refer to β_0 as the “RD gender gap,” since it measures the discontinuous change in the gender gap at the cutoff. When β_0 is not statistically different from zero, we cannot reject that the discontinuity in the outcome at the cutoff is the same for women and men. We report Eicker-White standard errors.

We restrict our sample to individuals with CEE scores close to the cutoff with an ad-hoc

¹⁰In RD designs with multiple cutoffs, having a unit exactly at the cutoff may challenge identification of average treatment effects. Fort, Ichino, Rettore, and Zanella (2022) show that adding cutoff fixed effects (intended-major fixed effects in our context) avoids this problem.

bandwidth of ± 59.89 points. This is the optimal bandwidth as chosen by the [Calonico, Cattaneo, Farrell, and Titiunik \(2019\)](#) procedure for a primary outcome of interest, enrollment in the 2020-1 admission cycle. Below each table we report the “Outcome mean above”, the average of the outcome of interest among applicants with a CEE score above the cutoff but within our ad-hoc bandwidth. We use this value to benchmark our results.

Our final analysis sample includes 4,287 applicants with CEE scores in the ad-hoc bandwidth. 2,861 (66%) submit at least one major to the slot allocation mechanism in the 2020-1 cycle and 1,308 (31%) reapplied at least once. 2,971 (69%) are enrolled in a major at Universidad Nacional within 4 admissions cycles. Finally, 3,988 (93%) are found in the SNIES Data attending a post-secondary education institution.

4.1 RD Validity

Identification requires applicants close to the intended-major admission cutoff to be comparable on both sides of the threshold. In particular, applicants should not be able to precisely manipulate their exam score relative to the cutoff. Importantly, whether the applicant’s score is above or below the cutoff depends on which major they chose to report as their intended major at exam registration, but they must report that intended major before they know their score. The identifying assumption is that observables (or other determinants of the outcome) evolve smoothly at the cutoff, while the outcome itself is expected to display a discontinuity if the treatment has an effect. We believe this assumption is plausible in our setting because applicants report their intended major before taking the exam, before learning their score, and before the current-year cutoffs are determined. Exam scores are graded by scantron and cannot be disputed or revised, making manipulation of the score unlikely. We nevertheless conduct several validity checks.

First, Appendix Figure [A2](#) plots the density of the running variable separately for women and men. Using the [Cattaneo, Jansson, and Ma \(2020\)](#) test, we fail to reject the null of no discontinuity in the density at the cutoff, with p-values of 0.1847 for women and 0.8173 for

men.

Second, we check that there is no differential selection into our linked dataset. Appendix Table A3 shows that there is no statistically significant gender gap discontinuity in the match rate across the three datasets we link.

Third, in Figure 1a and A3 we plot both the estimated coefficient on $Below_i$ and $Below_i \times Female_i$ from regressions placing each baseline characteristic on the left-hand-side to test for discontinuous jumps at the major cutoffs. We find little evidence of discontinuous changes in applicants' baseline characteristics at the cutoff nor are there differential jumps for marginal women compared to marginal men. The p-value for a test of joint significance for a gender gap discontinuity in covariates (excluding female and including indicators for missing values) using specification 1 is 0.94.

Finally, a potential concern for RD validity is that the intended major is self-reported and nonbinding and therefore even if the exam score is not manipulated, the running variable might be; for example, if an applicant could foresee what their score would be before they took the exam and what the yet-to-be-determined majors' admissions cutoffs would be, then they might report at registration an intended major they knew they would be admitted to. We test this possibility by looking to see if the likelihood of reporting a particular major as your intended major jumps at that major's cutoff, separately for each of the 49 majors. We plot the two estimated coefficients of interest with 95% confidence intervals from each of these 49 regressions in Figure 1b finding little evidence for manipulation of intended major reporting, even for majors with more plausibly anticipated cutoffs where the year-over-year change is small.¹¹ We then test the joint significance across all 49 regressions on the two coefficients of interest, and find no statistical evidence that the choice of intended major reported at registration is manipulated, i.e., no evidence that applicants whose score is right below a major's cutoff are discontinuously less likely to have reported that major as their intended major compared to applicants who score right above (p-value = 0.3692 on $Below_i$

¹¹In Appendix Table A4 we show there is no discontinuous change at the cutoff in the year-over-year change.

and 0.6660 on $Below_i \times Female$).

5 Results

5.1 Application outcomes

We first present results for immediate application outcomes, which occur before the current year’s admissions cutoffs are determined and published. Table 1 shows a substantial RD gender gap in immediate enrollment at Universidad Nacional (see Appendix Figure A4 for corresponding RD plot). Column 1 presents a gender-gap discontinuity of 14.2 pp in enrollment within the 24-hour priority-group window. The 14.2 pp RD gender gap corresponds to approximately 17% of the mean enrollment rate of 82% among applicants above the cutoff.

This immediate enrollment gap is accompanied by different use of the allocation mechanism by marginal applicants of either gender. Column 2 shows that the gender-gap discontinuity in the number of submitted majors within 24 hours is 0.165, marginally significant at the 10% level. The RD gender gap of 0.165 corresponds to approximately 11% of the mean of 1.49 submitted majors among applicants above the cutoff. Columns 3 and 4 examine whether the first and second major choice submissions are “safer” relative to their intended major, according to information available before the current cutoff is known. We define a submitted major as safe if the difference between the 2019 cutoff of the submitted major and the 2019 cutoff of the intended major is negative. We estimate an RD gender gap of 7.1 pp in submitting a safe first choice (significant at the 10% level), but we find no corresponding difference for the second choice. Panels (b) and (c) of Figure A4 show the same pattern visually.

Lastly, in Column 5, we show the effects of just missing the intended major’s admissions cutoff on retaking the exam in subsequent admission cycles. Retaking is an important alternative margin because reapplication requires taking the CEE again and only the new score counts in the new admission cycle. This outcome is mechanically related to the results from Column 1, immediate enrollment, as applicants are strongly discouraged from attempting

to switch majors once enrolled. We find that the discontinuity in retaking is 13.2 pp more negative for women than for men, indicating a smaller increase in retaking below the cutoff for women (Panel (e) in Figure A4).

5.2 Year-over-year cutoff changes

The fact that women seem to anticipate being just below the cutoff during the first 24-hour period seems puzzling since applicants should not know exactly what the cutoffs are until weeks after taking the exam. The results in Table 1 therefore cannot be interpreted as a reaction to an observed rejection. Instead, they suggest plausible anticipation of the cutoff around the eventual admission cutoff.

Figure 2 examines this possibility. We plot the estimated gender gap discontinuity (β_0) separately for each major on the y-axis against the absolute difference between the current year (2020-1 cycle) and last year’s (2019-1 cycle) cutoffs for that major. Applicants may plausibly anticipate being below the intended major’s cutoff if the cutoff did not change substantially. The plots exhibit a downward-sloping fitted line, suggesting that the gender gap discontinuities in both immediate enrollment (Panel (a)) and the number of submitted majors (Panel (b)) are larger when the year-over-year change is smaller. Panel (c) shows the same pattern for the outcome on submitting a safe first-choice major and Panel (d) for second choice majors. As expected, Panel (e) shows the opposite pattern for retaking as retaking is inversely related to enrollment in the first admissions cycle (Panel (a)).

Notably, for all the outcomes in Figure A4 that suggest a surprising “anticipatory response,” as the difference between the previous and current cutoffs increases, the estimates attenuate toward zero. This pattern is consistent with applicants only plausibly being able to “anticipate” having just missed or just made the admissions cutoff when the current year’s cutoff is similar to the last year’s cutoff. The concentration of these discontinuities among majors with small year-over-year cutoff changes is consistent with previous cutoffs being informative to applicants, but it does not explain why application behavior changes discon-

tinuously around the eventual cutoff. This heterogeneity analysis is presented as supportive rather than definitive evidence of anticipation. The year-over-year difference in cutoffs depends not only on randomness but also on the density of applicants near the margin and on how selective the university and its majors are overall. The exercise is considering a single factor, but other factors could be as important as the year-over-year change in the cutoff.

5.3 Downstream outcomes

Table 2 and Appendix Figure A5 of corresponding RD plots present results for outcomes downstream of the first admissions cycle, i.e., after the major cutoffs would be determined and published and after all applicants would have learned if they had scored above or below their intended major’s cutoff. Column 1 shows no statistically significant gender-gap discontinuity in enrollment at Universidad Nacional within two years. This suggests that the early enrollment gender gap largely attenuates over later admission cycles. Columns 2 and 3 use the SNIES Data containing the universe of post-secondary institution attendees in the whole country. The RD gender gap in enrollment at another university is -9.8 pp and is significant at the 10% level (Column 2). Following U. Nacional applicants with the SNIES, we also find no statistically significant RD gender gap in eventually enrolling in the intended major (Column 3), suggesting that despite the larger share of male marginal applicants retaking the exam, men are no more likely to ultimately enroll into their reported intended major compared to women who miss the same cutoff.

We next examine whether these differential responses are associated with different average earnings of the majors in which applicants enroll using the matched LMOE data as described in the Data Section 3. Table 2 Column 4 shows that, after four admission cycles, the gender gap in potential earnings changes discontinuously by 98,000 pesos at the cutoff, significant at the 5% level. This corresponds to a 4.9% gender gap in potential earnings among applicants below the cutoff. Column 5 shows a similar pattern when we look at enrollment at institutions outside Universidad Nacional. The gender gap in potential earnings changes discontinuously

by 128,000 pesos, implying a 4.5% gap below the cutoff. Using the earnings records of recent graduates, these results are meant to be descriptive of the types of majors men and women sort into and reflect field-level differences in early-career earnings, not the marginal effect of narrowly missing a cutoff. Since within-major earnings correlate with exam performance, this setup likely overstates the true difference at the admission margin. These results suggest that missing the admission cutoff can generate earnings-relevant sorting by gender.

The heterogeneity analysis in Figure 3 presents a different pattern than the one described so far for the immediate admissions outcomes. Panel (a) shows the gender gap for ever enrolled at U. Nacional by major, Panel (b) for ever enrolled in another university, Panel (c) for ever enrolled in one’s intended major, Panel (d) for potential earnings of the majors if ever enrolled at U. Nacional and Panel (e) for potential earnings of the majors if ever enrolled at another university. The scatterplots show that the gender gap discontinuity is negative for many majors, across the range of the cutoff absolute distance measure. This suggests that the potential earnings gap is not driven only by majors with smaller year-over-year cutoff variation. Rather, the RD gender gap in potential earnings remains negative across majors with both small and large year-over-year cutoff changes.

6 Robustness and Placebo Checks

We assess robustness to five alternative specifications in Figure A6 using a “waterfall plot.” The top point estimate and confidence interval present the baseline model. The remaining estimates assess sensitivity to five alternative specifications. We omit predetermined covariates, add controls for applicants’ first and second submitted choices to see if differences in submitted second choices drive our results,¹² control for applicants’ overall score rank to verify that the results are not driven by applicants with higher scores behaving differently from applicants with lower scores, use a quadratic specification in the running variable, and apply

¹²Because submitted choices are themselves outcomes and differ by gender around the cutoff, these estimates should not be interpreted causally.

triangular kernel weights. Across these specifications, the main patterns remain similar.

A second type of robustness checks is to test for sensitivity to the exclusion of majors where the puzzling anticipatory responses take place. Figure A7 sequentially excludes majors whose year-over-year cutoff change is below 5, 10, and 15 points. The immediate application discontinuities attenuate as majors with the most stable cutoffs are excluded, whereas the downstream estimates show no comparable gradient.

Figure A8 presents a similar coefficient waterfall plot at thresholds located in 5-point increments from 100 to 150 CEE score points below the realized intended-major cutoff. These placebo thresholds do not correspond to institutional admission margins and therefore should not generate discontinuous changes in the outcomes. The estimated gender-gap discontinuities are close to zero and not statistically significant in most cases, which supports the interpretation that the main estimates are specific to the actual admission cutoff.

7 Conclusion

A large literature explains gender differences in college major choice through differences in preferences, preparation, beliefs, competitiveness, and constraints. This paper shows that gender gaps can also emerge within the admissions process itself. Using applicants to Universidad Nacional who initially report intending the same major and score close to the same admission cutoff, we show that women and men have different outcomes. Women who narrowly miss the cutoff are more likely to secure admission earlier in an alternative major, while men are more likely to retake the entrance exam in later admission cycles.

Taken together, these findings highlight that institutional features of admissions, not just ex-ante field-of-study preference, shape the majors that applicants enroll in. Under plausibly anticipated but uncertain admission thresholds, next-best alternatives, retaking rules, and the information available about the cutoffs may generate earnings-relevant sorting even among applicants who initially target the same field.

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8 Tables

Table 1: Gender differences in application outcomes

	Enrolled	No. majors submitted	Submitted safe major		Any retake
	(1)	(2)	(3)	(4)	(5)
			First choice	Second choice	Cycles 2-4
Below	-0.481*** (0.029)	-0.012 (0.055)	0.048** (0.021)	0.081** (0.035)	0.283*** (0.029)
Below $\times$ Female	0.142*** (0.050)	0.165* (0.093)	0.071* (0.039)	0.001 (0.057)	-0.132*** (0.048)
Outcome mean above	0.82	1.49	0.05	0.32	0.09
Observations	4287	4287	4287	4287	4287

Notes: The table presents estimates of equation 1. *Below* is an indicator for scoring below the realized 2020-1 cutoff of the applicant's reported intended major, and *Below $\times$ Female* measures the differential discontinuity at the cutoff for women relative to men. Column 1 indicates enrollment at Universidad Nacional during the first 24-hour period in which the applicant can access the allocation mechanism. Column 2 reports the number of majors submitted during that period, ranging from zero to two. Columns 3 and 4 indicate whether the applicant submits a safe first or second choice, respectively, with zero also indicating if the applicants did not submit a major. A submitted major is defined as safe when its 2019-1 cutoff is lower than the 2019-1 cutoff of the applicant's reported intended major. Column 5 indicates whether the applicant retakes the CEE in any of the subsequent three admission cycles (2020-2 to 2021-2). The row "Outcome mean above" reports the mean of each outcome among applicants scoring above the intended-major cutoff within the ad hoc bandwidth equal to 59.89 points. All estimations include the controls and fixed effects described in equation 1, and report Eicker-Huber-White standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2: Gender differences in downstream enrollment and potential earnings

	Ever enrolled (within two years):			Potential earnings	
	(1) U. Nacional	(2) Other univ.	(3) In intended major	(4) U. Nacional	(5) Other inst.
Below	-0.092*** (0.027)	0.177*** (0.031)	-0.334*** (0.034)	-0.067** (0.030)	-0.021 (0.035)
Below $\times$ Female	0.020 (0.046)	-0.098* (0.053)	0.041 (0.056)	-0.098** (0.049)	-0.128** (0.060)
Outcome mean above	0.90	0.17	0.69	2.09	2.11
Observations	4287	4287	4287	2971	2426

Notes: The table presents estimates of equation 1 for outcomes measured after the first admission cycle. *Below* is an indicator for scoring below the realized 2020-1 cutoff of the applicant's reported intended major, and *Below $\times$ Female* measures the differential discontinuity at the cutoff for women relative to men. Columns 1-3 indicate whether, within four admission cycles (two years), the applicant enrolls at Universidad Nacional, enrolls at another university, or enrolls in the reported intended major at any university, including Universidad Nacional. Column 1 uses Universidad Nacional admissions records, while Columns 2 and 3 use the SNIES administrative enrollment data. Columns 4 and 5 report the potential earnings associated with the institution-major pair in which the applicant enrolls at Universidad Nacional or at another institution, respectively. Potential earnings are measured in millions of Colombian pesos per month and are based on the average earnings of recent graduates from the corresponding institution-major pair in the LMOE data. The row "Outcome mean above" reports the mean of each outcome among applicants scoring above the intended-major cutoff within the relevant estimation sample and the ad hoc bandwidth equal to 59.89 points. All estimations include the controls and fixed effects described in equation 1, and report Eicker-Huber-White standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

9 Figures

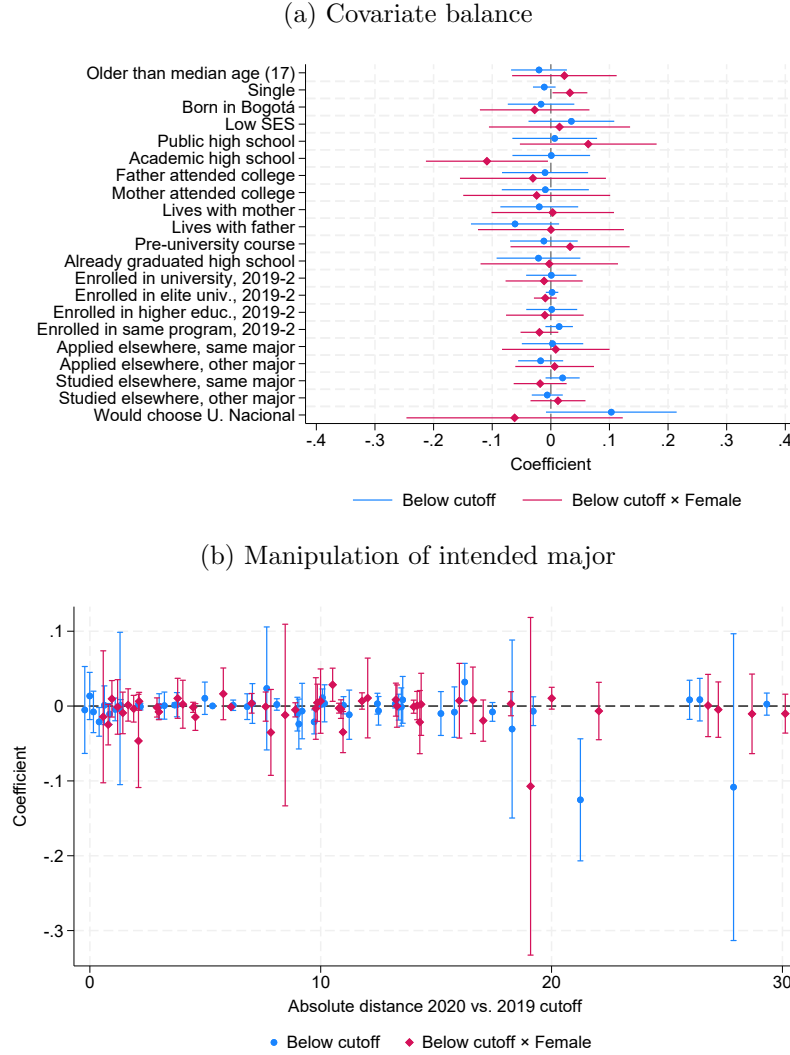


Figure 1: RD validity: covariate balance and intended-major reporting

Notes: Panel (a) reports estimates of equation 1 using each predetermined applicant characteristic listed on the y-axis as the dependent variable. For each characteristic, the figure plots the coefficient on *Below* and the coefficient on *Below $\times$ Female*. Horizontal bars show 95% confidence intervals based on Eicker-Huber-White standard errors. Missing values are imputed as zero to keep the sample size constant and the analogous plot for the missing indicators is shown in Figure A3. Panel (b) examines whether reporting a particular intended major changes discontinuously at that major's realized cutoff. The specification is estimated separately for each of the 49 intended majors, and the coefficients on *Below* and *Below $\times$ Female* are plotted against the absolute difference between that major's 2020-1 and 2019-1 cutoffs. Vertical bars show 95% confidence intervals.

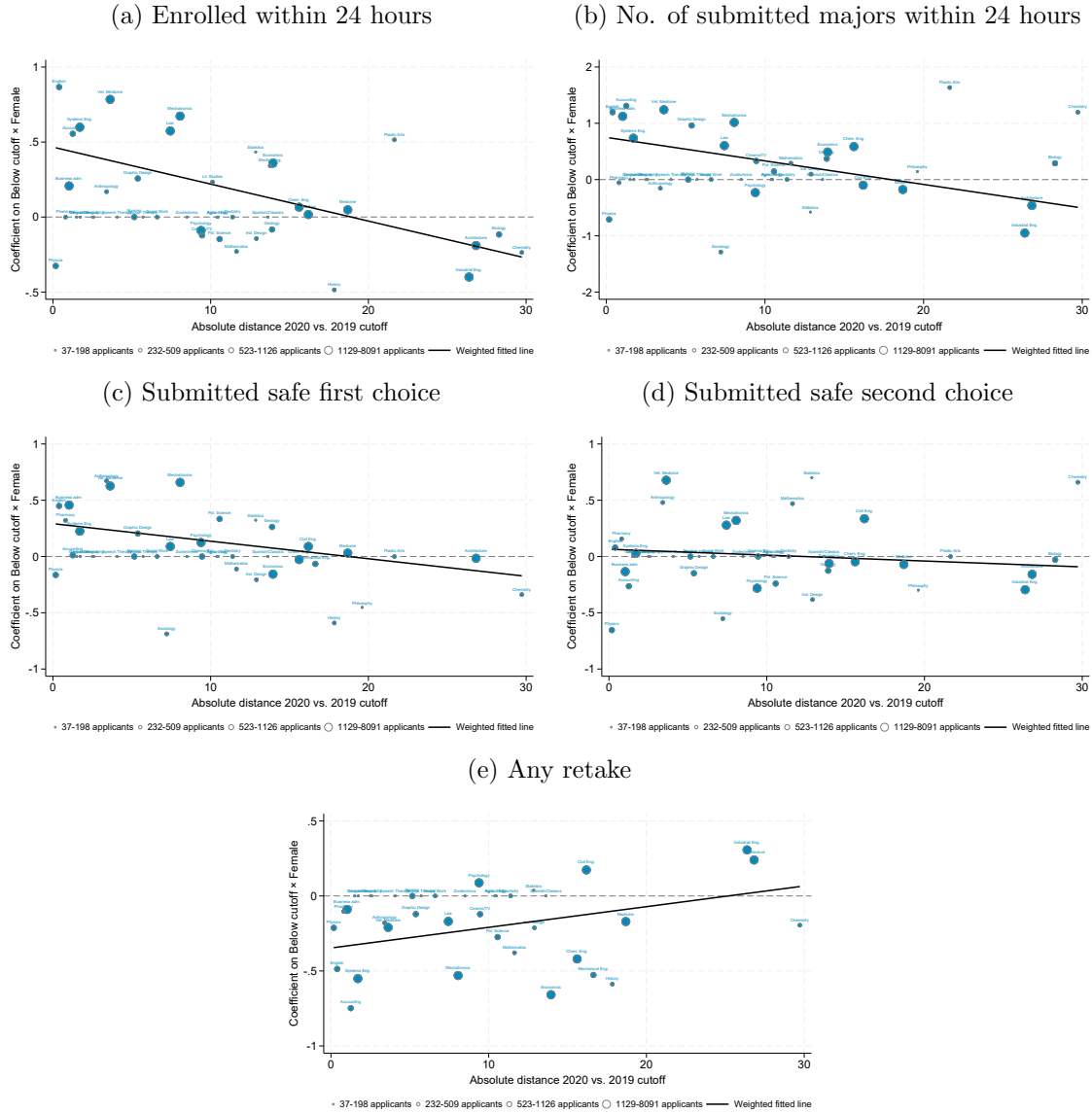


Figure 2: Heterogeneity in application outcomes by year-over-year cutoff change

Notes: Each point represents one reported intended major and plots the estimated coefficient on $Below \times Female$ from equation 1, estimated separately within that major. The coefficient measures the differential discontinuity at the intended-major cutoff for women relative to men. The x-axis reports the absolute difference between the major's 2020-1 and 2019-1 admission cutoffs. Panels (a)–(e) report estimates for enrollment within 24 hours, number of majors submitted within 24 hours, submission of a safe first choice, submission of a safe second choice, and retaking the CEE, respectively. Marker size reflects the number of applicants reporting each intended major. The solid line reports the fitted relationship across majors, weighted by the number of applicants in each intended major.

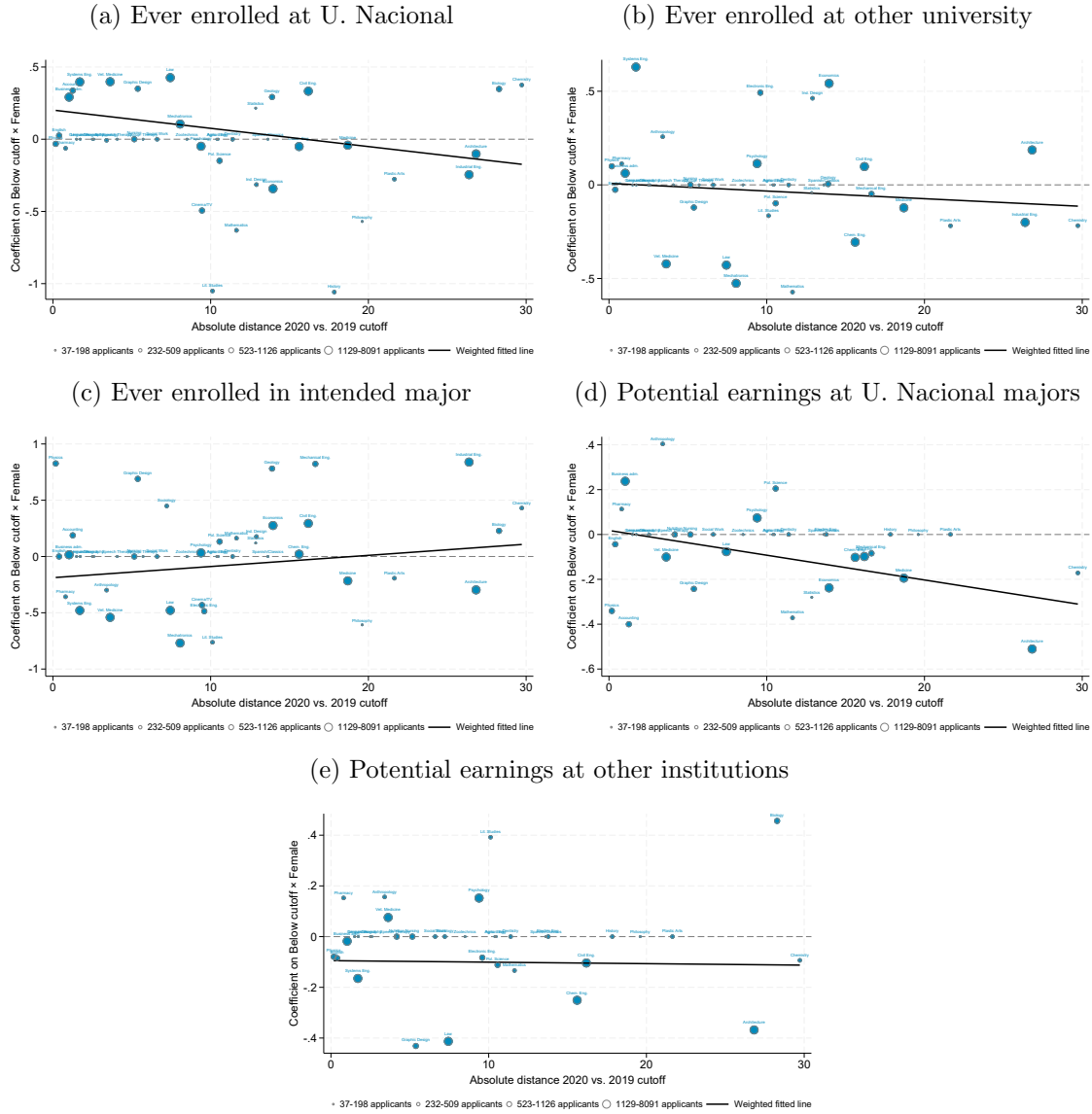


Figure 3: Heterogeneity in downstream outcomes by year-over-year cutoff change

Notes: Each point represents one reported intended major and plots the estimated coefficient on *Below* $\times$ *Female* from equation 1, estimated separately within that major. The coefficient measures the differential discontinuity at the intended-major cutoff for women relative to men. The x-axis reports the absolute difference between the major's 2020-1 and 2019-1 admission cutoffs. Panels (a)–(c) report whether applicants enroll within two years at Universidad Nacional, at another university, or in their reported intended major at any university, respectively. Panels (d) and (e) report the potential earnings associated with the institution-major pair in which applicants enroll at Universidad Nacional and at other institutions, respectively. Marker size reflects the number of applicants reporting each intended major. The solid line reports the fitted relationship across majors, weighted by the number of applicants in each intended major.

10 Appendix tables

Table A1: Major-specific admission cutoffs and available slots, 2019-1 and 2020-1 (Part 1)

College major	Cutoff 2019-1	No. slots 2019-1	Cutoff 2020-1	No. slots 2020-1
Agricultural Engineering	637.8989	106	627.5051	105
Agronomic Engineering	635.624	90	625.1466	100
Anthropology	662.0044	35	658.6011	35
Architecture	694.9775	66	668.1536	82
Biology	728.266	30	699.9824	48
Business Administration	652.3063	120	653.3345	70
Chemical Engineering	687.6607	129	672.054	134
Chemistry	700.1125	75	670.3881	69
Cinema and Television	613.6038	28	623.0564	28
Civil Engineering	694.696	109	678.5037	109
Computer and Systems Engineering	700.5173	100	698.8107	100
Computer Science	707.0986	33	707.6533	28
Dentistry	638.5678	75	627.1788	90
Economics	681.2656	120	667.314	105
Electric Engineering	690.7929	55	677.0153	60
Electronic Engineering	721.8709	60	712.2769	60
Geography	629.0294	35	626.4404	32
Geology	706.6311	33	692.7388	33
Graphic Design	678.183	32	683.5642	32
History	645.4142	60	627.5738	60
Industrial Design	672.6489	50	659.7461	46
Industrial Engineering	710.3484	44	683.9687	44
Law	688.647	82	681.2057	80
Linguistics	626.2433	45	624.5093	50
Literary Studies	650.4933	43	660.6105	44

Notes: The table reports admission cutoffs and the predetermined number of available slots for each major at the Bogotá campus in admission cycles 2019-1 and 2020-1. The 2020-1 cycle is the admission cycle studied in the paper, while 2019-1 is the corresponding cycle one year earlier. A major's cutoff is the CEE score of the applicant obtaining the last available slot in that major. Cutoffs are realized during the allocation process, whereas the number of slots is determined by the University before the admission cycle. Music programs, which have additional admission requirements beyond the CEE, are excluded. Admission statistics obtained from: <https://admisiones.unal.edu.co/servicios-en-linea/estadisticas-del-proceso-de-admision/>.

Table A2: Major-specific admission cutoffs and available slots, 2019-1 and 2020-1 (Part 2)

College major	Cutoff 2019-1	No. slots 2019-1	Cutoff 2020-1	No. slots 2020-1
Mathematics	713.0775	45	724.7126	32
Mechanical Engineering	702.5432	81	685.9037	81
Mechatronics Engineering	775.8461	45	767.7845	45
Medicine	776.6215	122	757.9305	128
Nursing	643.9633	89	638.8036	79
Nutrition and Diet	652.4565	50	648.2854	46
Occupational Therapy	625.2646	60	619.5499	55
Pharmacy	682.9627	56	683.7646	54
Philology and Languages: English	652.0954	44	651.7011	43
Philology and Languages: French	649.1097	22	646.6024	22
Philology and Languages: German	648.5624	21	650.06	22
Philosophy	645.7763	50	626.1694	50
Physics	726.0368	73	725.8575	89
Plastic Arts	616.9134	25	595.265	32
Political Science	657.8519	90	647.2855	86
Psychology	663.7908	75	673.1798	60
Public Accounting	637.2641	128	638.5212	100
Social Work	637.7145	41	631.1059	42
Sociology	653.783	39	646.568	45
Spanish and Classical Philology	637.3839	40	623.7555	44
Speech Therapy	625.3862	75	621.3111	60
Statistics	681.5716	55	668.7146	47
Veterinary Medicine	658.7451	50	655.1168	50
Zootechnics	629.7345	55	621.2348	60

Notes: The table reports admission cutoffs and the predetermined number of available slots for each major at the Bogotá campus in admission cycles 2019-1 and 2020-1. The 2020-1 cycle is the admission cycle studied in the paper, while 2019-1 is the corresponding cycle one year earlier. A major's cutoff is the CEE score of the applicant obtaining the last available slot in that major. Cutoffs are realized during the allocation process, whereas the number of slots is determined by the University before the admission cycle. Music programs, which have additional admission requirements beyond the CEE, are excluded. Admission statistics obtained from: <https://admisiones.unal.edu.co/servicios-en-linea/estadisticas-del-proceso-de-admision/>.

Table A3: RD estimates for administrative-data match rates

	(1)	(2)
	Applicant in SNIES data	Program and institution matched to LMOE data
Below	-0.010 (0.018)	-0.012 (0.023)
Below $\times$ Female	0.003 (0.027)	-0.016 (0.034)
Outcome mean above	0.95	0.93
Observations	4287	4287

Notes: The table presents estimates of equation 1 for indicators of successful linkage across administrative datasets. Column 1 indicates whether an applicant in the Universidad Nacional Admissions Data is observed attending a higher-education institution in the SNIES data between 2020-1 and 2021-2 (within four admission cycles or two years). Column 2 indicates whether the institution-major pair in which the applicant enrolls can be matched to potential-earnings information in the LMOE data. *Below* is an indicator for scoring below the realized 2020-1 cutoff of the applicant’s reported intended major, and *Below $\times$ Female* measures the differential discontinuity at the cutoff for women relative to men. The row “Outcome mean above” reports the mean of each outcome among applicants scoring above the intended-major cutoff within the ad hoc bandwidth equal to 59.89 points. All estimations include the controls and fixed effects described in equation 1, and report Eicker-White standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A4: RD balance in year-over-year cutoff change

	(1) Absolute distance (2019-2020)
Below	0.115 (0.625)
Below $\times$ Female	-0.141 (1.043)
Outcome mean above	11.55
Observations	4287

Notes: The dependent variable is the absolute difference between the 2020-1 and 2019-1 admission cutoffs of the applicant's reported intended major at Universidad Nacional. *Below* is an indicator for scoring below the realized 2020-1 cutoff, and *Below $\times$ Female* measures the differential discontinuity at the cutoff for women relative to men. The regression includes priority group fixed effects and baseline covariates. The row "Outcome mean above" reports the mean absolute year-over-year cutoff difference among applicants scoring above the intended-major cutoff within the ad hoc bandwidth equal to 59.89 points. Eicker-Huber-White standard errors are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A5: Major submissions and slot availability by priority group

		Applicants in priority group:			
		1	2	3	4
Panel A. Major selection and slot allocation					
24h period Majors submitted					
1	0	180			
	1	389			
	2	920			
2	0	308	532		
	1	4	335		
	2	29	1,200		
3	0	308	1,004	971	
	1	0	20	139	
	2	0	87	570	
4	0	308	1,002	1,134	1,692
	1	0	1	9	101
	2	0	1	37	543
Panel B. No. of slots and applicants by priority group					
Majors with slots available		49	43	20	7
Total applicants in priority group		1,489	2,067	1,680	2,336
% women		30.6	34.4	36.1	40.2
% men		69.4	65.6	63.9	59.8
Not enrolled at the end of cycle 1		308	1,002	1,135	2,262
% of total women		22.4	45.2	65.6	95.7
% of total men		19.9	50.2	68.7	97.6

Notes: Panel A reports the number of applicants submitting zero, one, or two major choices in each 24-hour allocation period, separately by the applicant's original priority group. Applicants who are not assigned a slot during their first 24-hour period may participate again with a subsequent priority group if slots remain available. Panel B reports the number of majors with available slots at the beginning of each priority group's allocation period, the total number of applicants assigned to each priority group, their gender composition, and the number and share not enrolled at the end of the first admission cycle. The 2020-1 Bogotá allocation mechanism contains 49 majors in the analysis sample after excluding music programs, which have additional admission requirements beyond the CEE.

11 Appendix figures

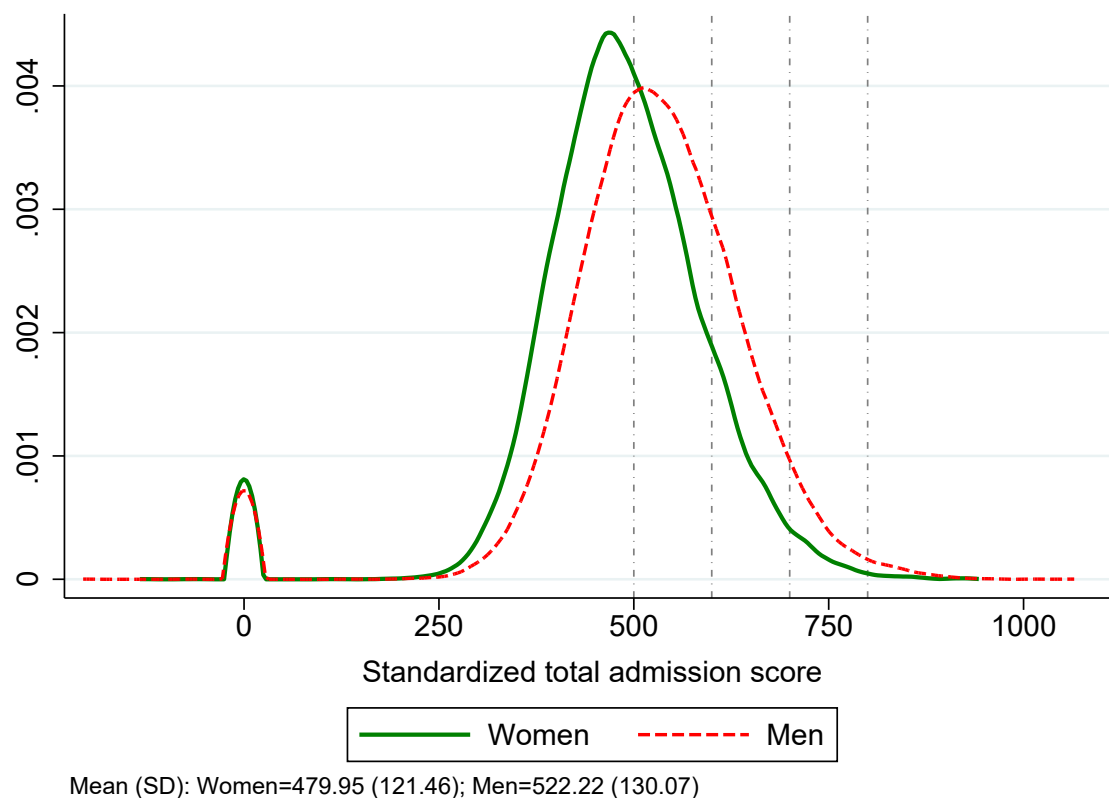


Figure A1: Distribution of CEE scores by gender

Notes: The figure plots the distribution of standardized CEE scores for women and men among applicants to the 2020-1 admission cycle. The CEE is standardized to have a mean of 500 and a standard deviation of 100. The vertical reference lines indicate scores of 500, 600, 700, and 800. The mass around zero corresponds primarily to registered applicants who did not take the exam. Mean scores and standard deviations by gender are reported below the figure.

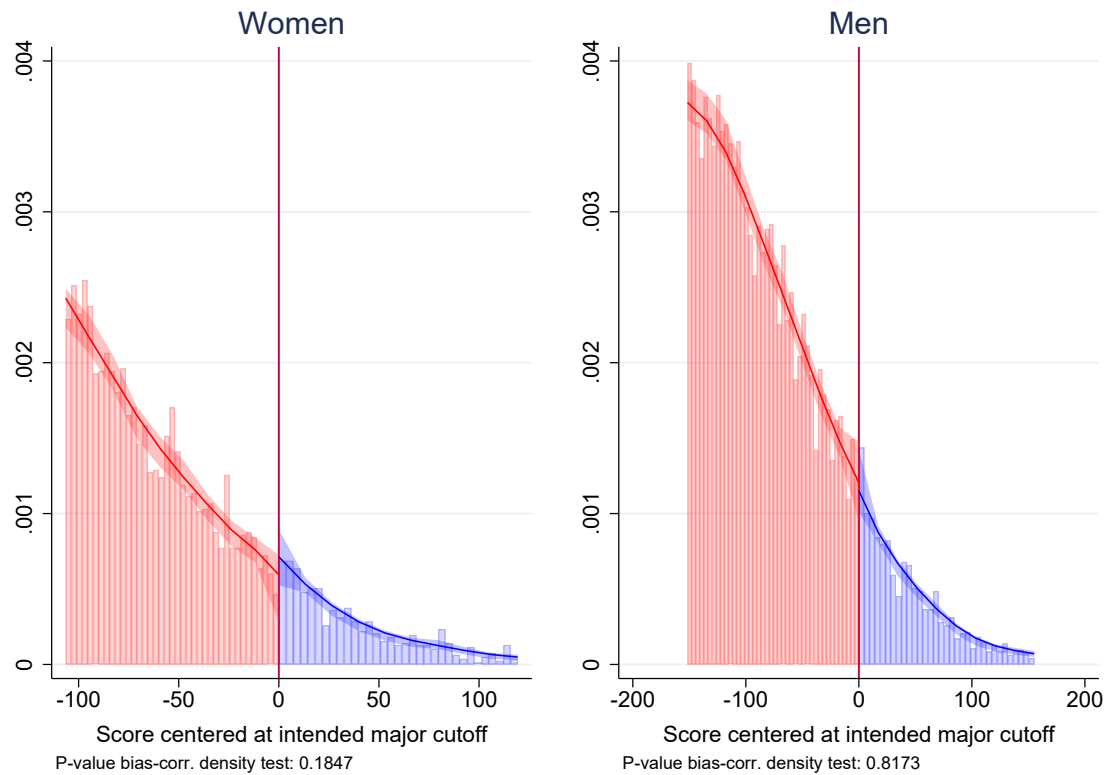


Figure A2: Density of the running variable by gender

Notes: The figure plots the density of the running variable separately for women and men. The running variable is the applicant's CEE score centered at the realized cutoff of the reported intended major, such that zero corresponds to the intended-major cutoff. The figure reports bias-corrected density discontinuity tests separately by gender. The vertical line indicates the cutoff.

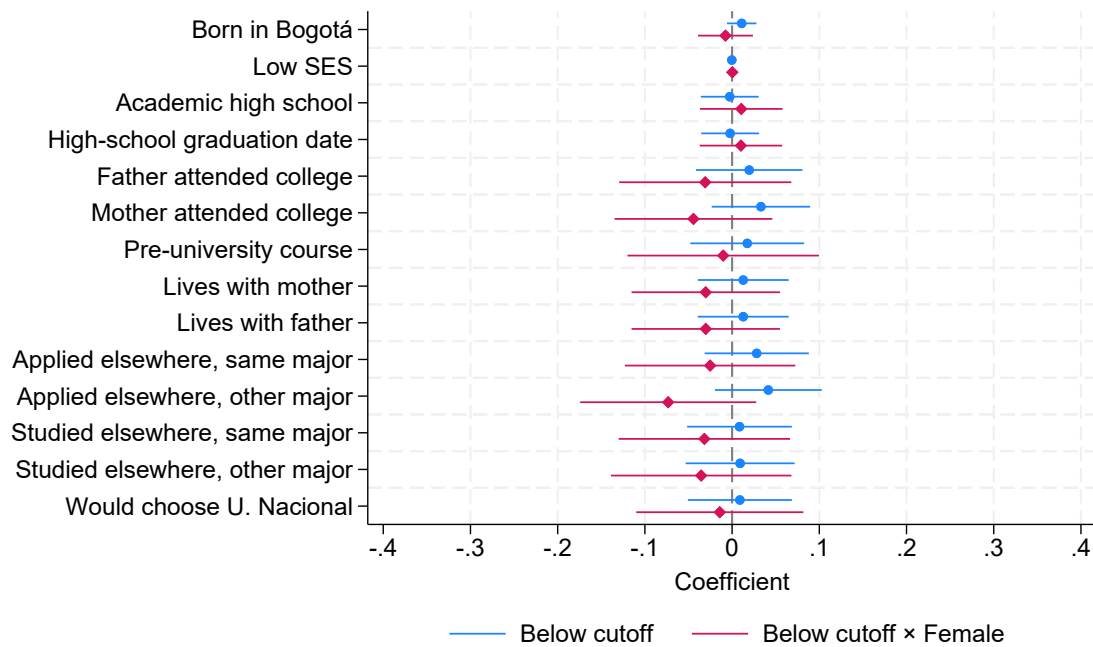


Figure A3: RD validity: balance of missing indicators for covariates

Notes: For each missing indicator, the figure plots the coefficient on *Below* and the coefficient on *Below × Female*. Horizontal bars show 95% confidence intervals based on Eicker-Huber-White standard errors.

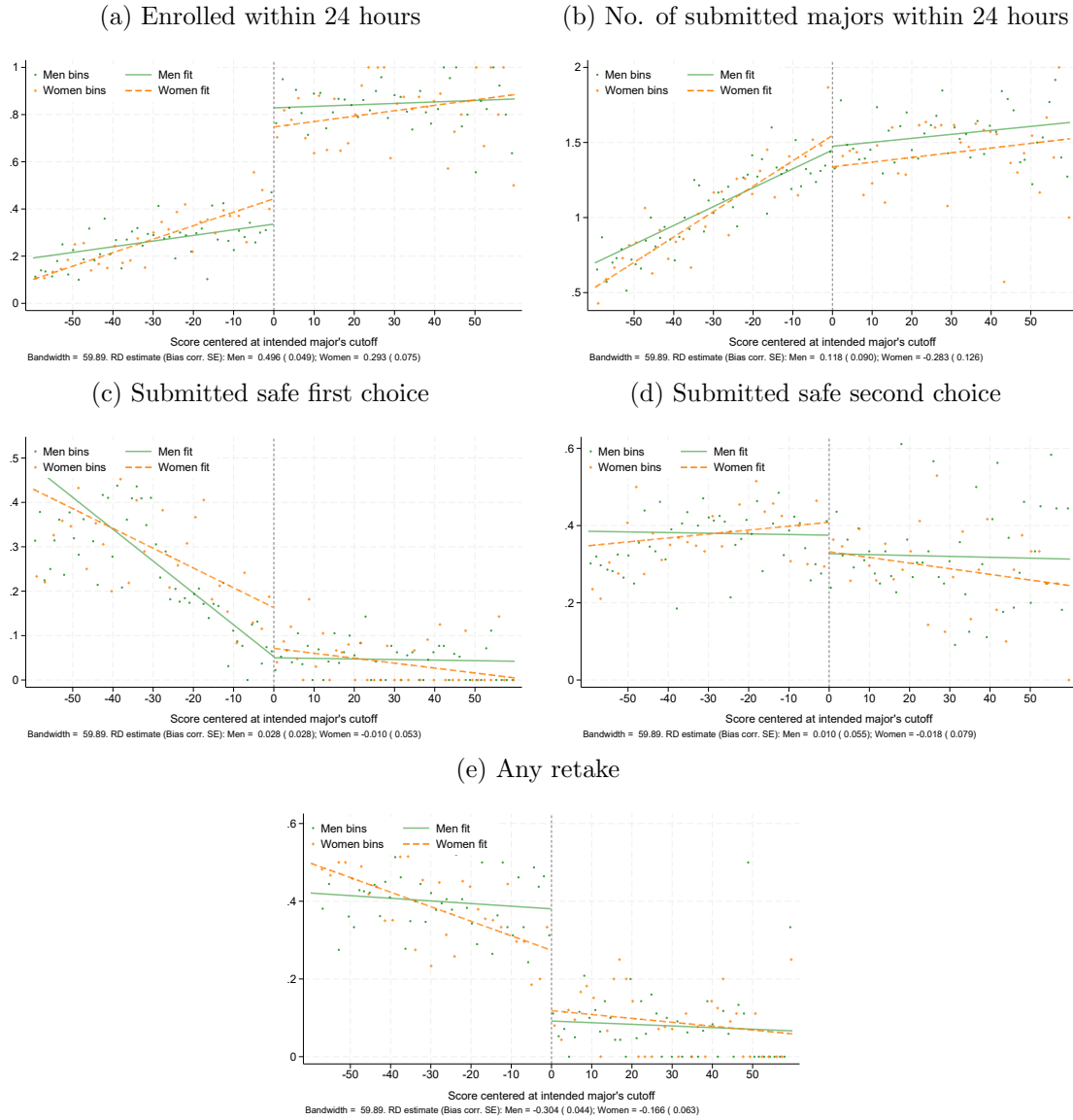


Figure A4: RD plots for application outcomes

Notes: The figure presents regression-discontinuity plots separately for women and men for the immediate admission outcomes reported in Table 1. The running variable is the applicant's CEE score centered at the realized cutoff of the reported intended major. Points show binned outcome means and lines show fitted values on either side of the cutoff separately by gender. All panels use the ad hoc bandwidth equal to 59.89 points. Bias-corrected gender-specific RD estimates and standard errors based on [Calonico, Cattaneo, and Titiunik \(2014\)](#) are reported below each panel. Panels (a)–(e) correspond to enrollment within 24 hours, number of majors submitted within 24 hours, submission of a safe first choice, submission of a safe second choice, and retaking the CEE, respectively.

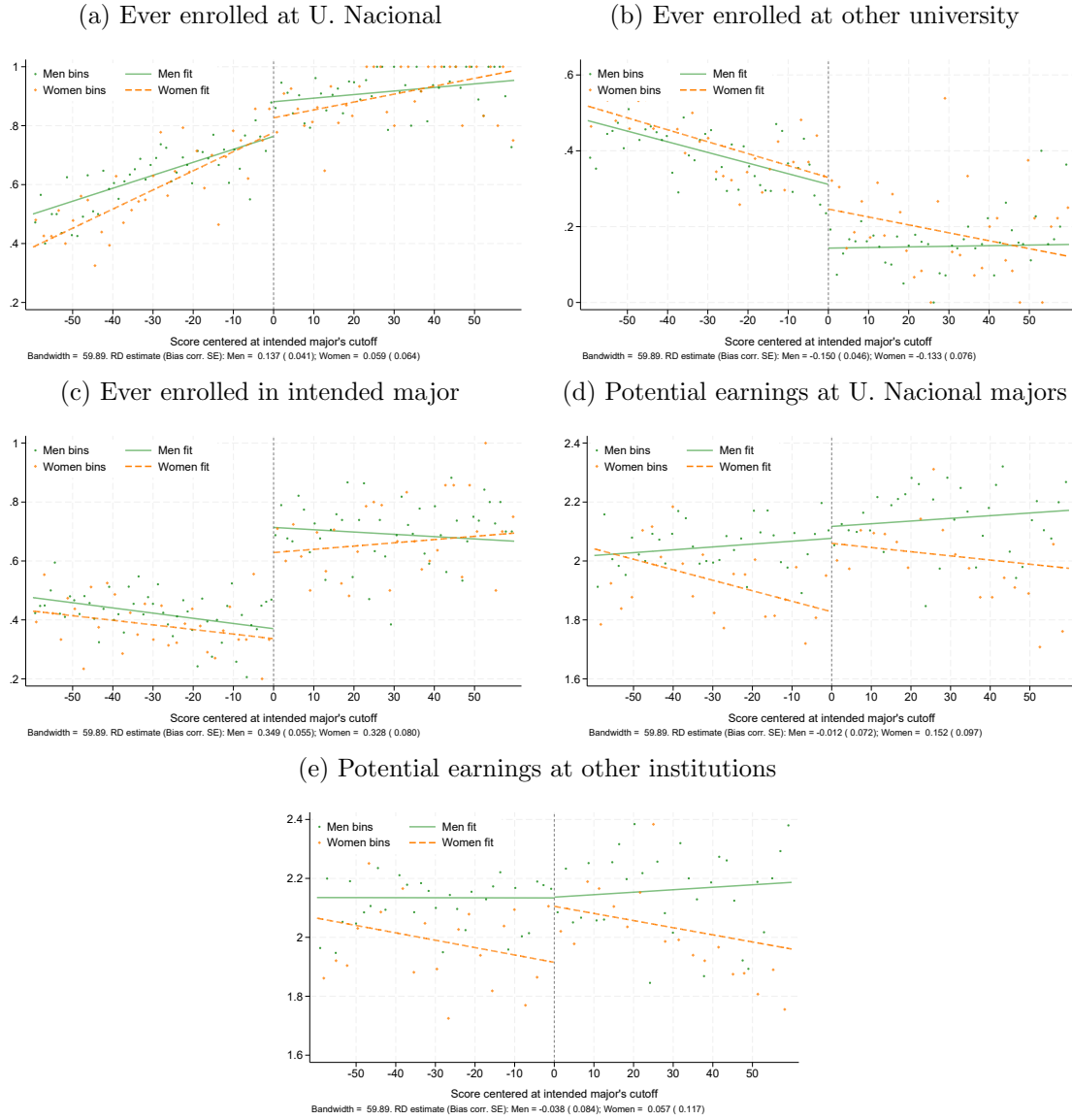


Figure A5: RD plots for downstream outcomes

Notes: The figure presents regression-discontinuity plots separately for women and men for the downstream outcomes reported in Table 2. The running variable is the applicant's CEE score centered at the realized cutoff of the reported intended major. Points show binned outcome means and lines show fitted values on either side of the cutoff separately by gender. All panels use the ad hoc bandwidth equal to 59.89 points. Bias-corrected gender-specific RD estimates and standard errors based on [Calonico et al. \(2014\)](#) are reported below each panel. Panels (a)–(c) report enrollment within two years at Universidad Nacional, at another university, and in the reported intended major at any university. Panels (d) and (e) report potential earnings associated with majors at Universidad Nacional and at other institutions, measured in millions of Colombian pesos per month.

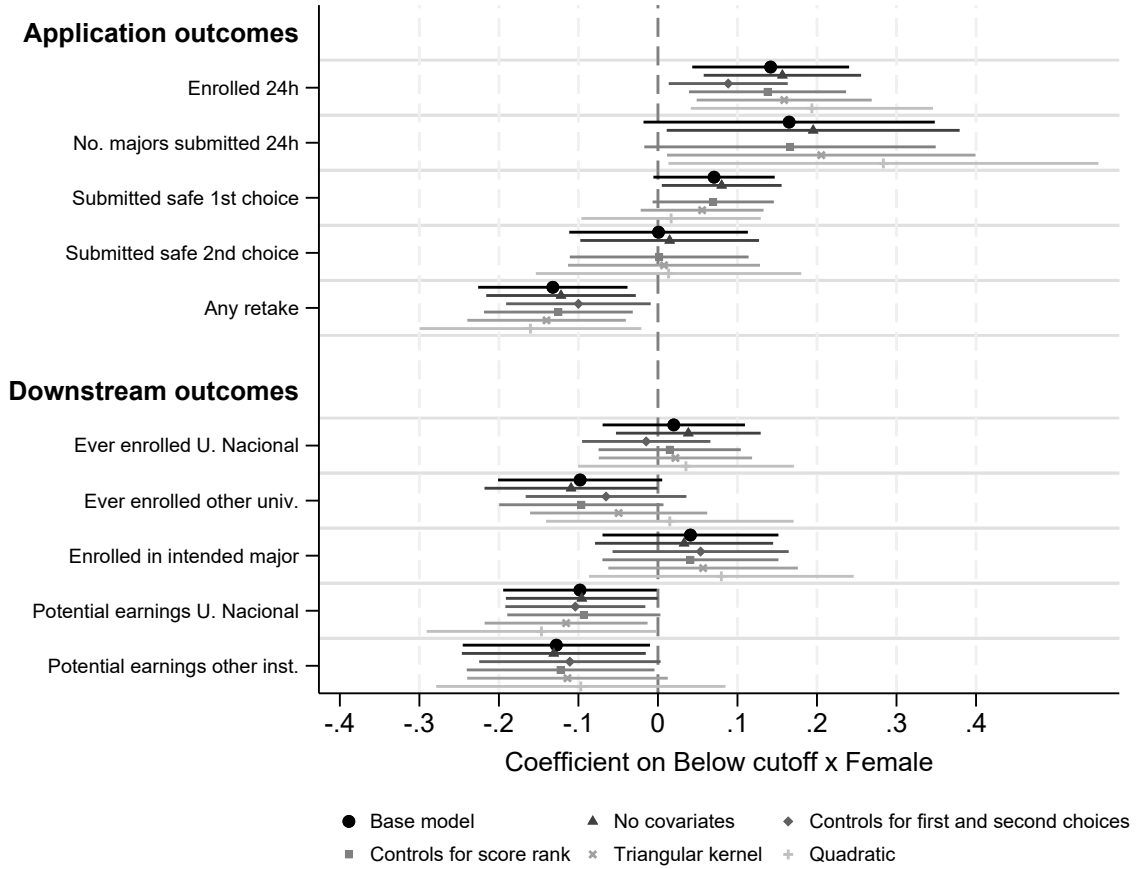


Figure A6: Robustness to alternative RD specifications

Notes: The figure plots the estimated coefficient on $Below \times Female$ from equation 1 for each main outcome. The coefficient measures the differential discontinuity at the intended-major cutoff for women relative to men. The baseline specification is compared with five alternatives: omitting predetermined covariates; adding controls for applicants' first and second submitted major choices, where applicable; controlling for applicants' overall CEE score rank; applying a triangular kernel that gives greater weight to observations closer to the cutoff; and replacing the baseline local linear specification in the running variable with a quadratic specification. All estimations use the ad hoc bandwidth equal to 59.89 points and otherwise include the controls and fixed effects described in equation 1, except where modified by the robustness check. Horizontal bars show 95% confidence intervals based on Eicker-Huber-White standard errors. Potential earnings are measured in millions of Colombian pesos per month.

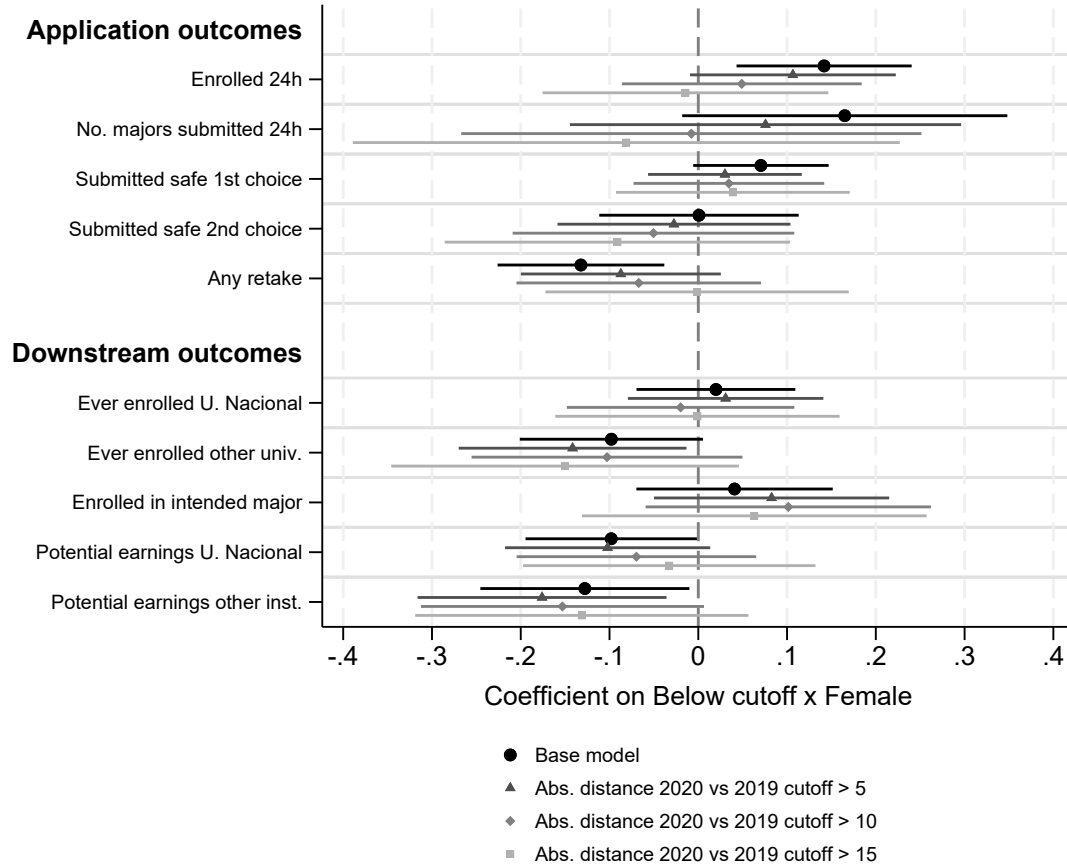


Figure A7: Sensitivity to excluding majors with stable cutoffs

Notes: The figure plots the estimated coefficient on $Below \times Female$ from equation 1 for each main outcome. The coefficient measures the differential discontinuity at the intended-major cutoff for women relative to men. The black point reports the baseline estimate using all intended majors. The remaining estimates sequentially restrict the sample to majors whose absolute difference between the 2020-1 and 2019-1 admission cutoffs exceeds 5, 10, and 15 score points, respectively. These restrictions progressively exclude majors with the smallest year-over-year cutoff changes. All estimations use the ad hoc bandwidth equal to 59.89 points and otherwise include the controls and fixed effects described in equation 1. Horizontal bars show 95% confidence intervals based on Eicker-White standard errors. Potential earnings are measured in millions of Colombian pesos per month.

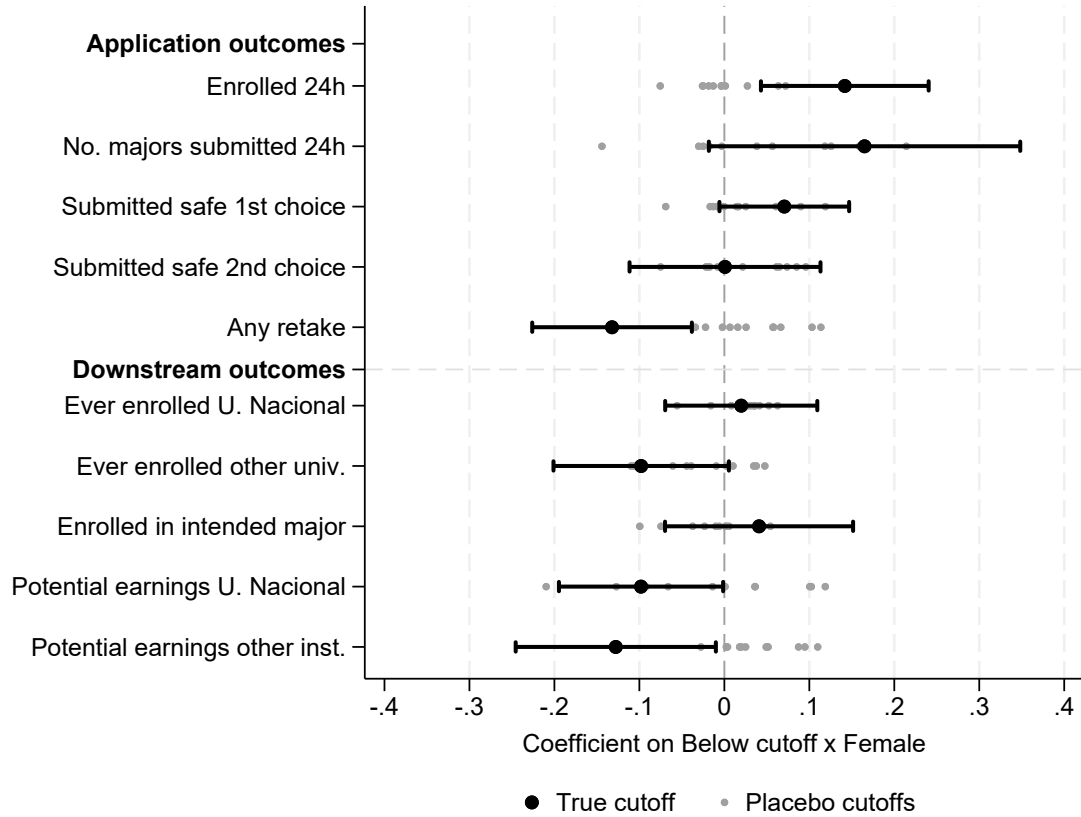


Figure A8: Placebo-cutoff tests

Notes: The figure compares the estimated coefficient on $Below \times Female$ at the realized intended-major cutoff with estimates obtained at placebo cutoffs located in 5-point increments from 100 to 150 CEE score points below the realized cutoff. The coefficient measures the differential discontinuity for women relative to men. Black points report estimates at the realized cutoff and gray points report estimates at the 11 placebo cutoffs. The placebo cutoffs do not correspond to institutional admission margins. Each placebo estimation recenters the running variable at the corresponding placebo cutoff and otherwise includes the controls and fixed effects described in equation 1. All estimations use the ad hoc bandwidth equal to 59.89 points around the corresponding cutoff. Horizontal bars for the realized-cutoff estimates show 95% confidence intervals based on Eicker-White standard errors. Potential earnings are measured in millions of Colombian pesos per month.

A Institutional Background

A.1 Higher education in Colombia and Universidad Nacional

The higher education system in Colombia consists of public and private institutions, with the latter outnumbering the former by a ratio of 2.6.¹³ College admissions are decentralized and using exams as part of the admissions criteria is widespread. Universities use the national standardized high school exam (prueba SABER) or their own entrance exam. Public universities subsidize tuition according to the household income of the student. Universidad Nacional de Colombia is the largest and most prestigious public university in Colombia with 9 branch campuses.¹⁴ Our study focuses on applicants to the Bogotá campus, which offers highly sought-after majors such as medicine and attracts nearly 70% of all applicants, according to our tabulations.

A.2 College Entrance Exam (CEE) and Admissions

Universidad Nacional de Colombia has two admission cycles per year. The university gives the CEE in March for admission the following August, and in September for admission the following January. We study the admission process for applicants who took the CEE in September 2019 for admission in January 2020 (2020-1 or first admission cycle henceforth).

Because the university requires no other application materials besides the CEE,¹⁵ we call all CEE takers “applicants.” Up to one month before the CEE takes place, applicants pay the exam fee, which is about \$25 USD, and register for the exam. Applicants report the branch campus they are applying to and their intended major on the exam registration form. Their intended major is nonbinding as applicants can change their minds and submit any available

¹³There are 83 public and 213 private institutions including vocational, and “university foundations” (fundaciones/corporaciones universitarias in Spanish) <https://www.universidad.edu.co/ya-va-en-296-el-numero-de-ies-en-colombia/>.

¹⁴Four branch campuses are located in the Andean region (Bogotá, Medellín, Manizales and Palmira), one in Cesar (La Paz) and four National Presence Campuses, located in the Amazon, Orinoco, Tumaco and San Andrés.

¹⁵Applicants to music majors must also submit examples of their work and for this reason we exclude intended music majors for our study.

major to the slot allocation mechanism. The intended major is also unconstrained as the student can choose any major the university offers. The reported intended major plays no role in admissions decisions and is collected solely to inform administrators about the relative demand for different fields. After registering for the CEE, the university informs applicants of the time, date and an individually assigned location to take the CEE in person, on paper. All applicants take the same CEE regardless of their intended major. No other university uses this university’s CEE score nor has access to the questions in the CEE. Because the reported intended major plays no role in the allocation mechanism and is reported before applicants know their exam score, we treat it as informative about applicants’ ex-ante intended field, while not assuming that it represents a firm commitment or a binding preference ranking.

The exam lasts 3.5 hours and contains multiple choice questions in mathematics (25 questions), natural sciences (25 questions), social sciences (25 questions), text analysis (25 questions), and image analysis (20 questions), for a total of 120 questions. Students receive no penalty for wrong answers. A scantron machine grades the exam and scores cannot be disputed or revised. Scores can therefore not be manipulated ex-post nor can the grading be affected by direct discrimination. The final score, a sum of all exam components, is standardized with a mean of 500 and a standard deviation of 100. The standardized score ranges from 0 to 1,000 points (see Appendix Figure A1). The University releases the exam results within two weeks. At that time, applicants learn their individual score for each exam component, their overall score, their score rank compared to all applicants and compared to applicants to the same branch campus, and their admission priority group. The admission priority group gives the order in which applicants can access an online platform to submit majors for consideration by the slot allocation mechanism.

A.3 College Major Choice Submission

The University assigns applicants who score at least 600 points (1 SD above the mean) into four admission priority groups based on their CEE score. This group determines the

applicant’s level of priority to submit major choices on the University’s online platform. In the Bogotá Campus, applicants in group 1 score at least 700 (2 SDs above the mean); group 2 applicants have scores between 650 and 700, group 3 applicants between 625 and 650, and group 4 applicants between 600 and 625. Each priority group is assigned a 24-hour period to log into the online platform to submit up to two majors indicating their first and second ranked choice.

When applicants log into the online platform, they see the majors with slots available,¹⁶ and know their CEE score and their admission priority group. Applicants *may* know the major-specific cutoff scores from previous admission cycles from published admission statistics. Applicants do not know how many other applicants are in their priority group, which majors other applicants in their group will submit, or which CEE score will be the cutoff score for any of the majors in the current admission cycle.

After seeing their set of available major choices, applicants must decide first whether or not to submit any majors at all. Applicants who submit no majors drop out of the admissions process.¹⁷ Applicants who decide not to drop out of the process may submit up to two majors to be considered for enrollment. Among applicants who submit at least one major, 78% of male applicants and 75% of female applicants submit two majors within the first 24-hour period they can access the platform.

Once the 24-hour period ends, the University closes the online platform and the slot allocation mechanism assigns available slots in each major. We describe the slot allocation mechanism next.

¹⁶Applicants in priority group 1 see all the majors, but each subsequent priority group only sees the majors available at the beginning of their 24-hour period. The platform does not provide the number of slots left for each major in real time, but instead the list of majors with slots available.

¹⁷Table A5 shows that 44.4% of applicants assigned a priority group do not submit any majors in the online platform. Nearly 80% of applicants dropping out are in priority groups 3 and 4, when very few majors are left. A minority of applicants in groups 1 and 2 do not submit any majors. These applicants may be studying at another university, seeking a slot at another institution, taking a test prep course, studying to retake the CEE, working or in the military, etc.

A.4 Slot Allocation Mechanism

The University’s admissions process aims to assign slots to the applicants with the top CEE scores. The University uses a Serial Dictatorship slot allocation mechanism with a truncated choice list and sequential slot assignment over four stages (one for each priority group). There are 49 total majors offered at the University.¹⁸ Each major has a predetermined number of slots to be assigned. The international survey of tertiary admission systems by [Immorlica, Leshno, Lo, and Lucier \(2020\)](#) shows that these features are not unique to Colombia. Out of the 30 systems they document, 11 use decentralized admissions, 4 use truncated or restricted choice lists, and 3 use priority groups or multi-stage sequential implementations. Our setting thus shares key features with admission processes worldwide while also offering unusually rich data. Specifically, we observe applicants’ unconstrained reported intended majors—recorded before receiving exam scores—the major choices they subsequently submit to the allocation mechanism, and their realized enrollments. This allows us to study how applicants’ ex-ante reported intentions relate to subsequent choices and realized enrollment in a non-strategy-proof mechanism.

Within priority group, the mechanism ranks all applicants from highest to lowest CEE score. Then the mechanism assigns the top scoring applicant their first choice major. Then the mechanism moves on to the next highest scoring applicant. The mechanism assigns that applicant to their first choice major if a slot is still available and, if not, assigns them to their second choice major if a slot is still available. The mechanism moves on to the next highest scoring applicant and so on and so forth. An applicant could be assigned a slot to their second choice major even if a lower scoring student had chosen this major as their first choice. Applicants who are not assigned a slot for their first or second major choice are not enrolled in that 24-hour period.

After the mechanism considers the choices of each applicant in the priority group, the

¹⁸In the admission cycle we study, there are 49 available majors, excluding the music majors which have additional admission requirements beyond the CEE.

University updates the set of available majors shown to the next priority group in the online platform. If the allocation mechanism does not assign an applicant to a slot, she has the option of logging into the next 24-hour period (i.e., with the next priority group) and submitting up to two new major choices. Alternatively, she can drop out of the current admission cycle. Therefore, it is possible to observe applicants submitting more than two major choices if they decide to submit majors in more than one 24-hour period. However, most applicants do not submit more than once.¹⁹

Table A5 shows how many applicants per admission priority group submitted 0, 1 and 2 majors in every 24-hour period. Applicants who decide to submit majors with the next priority group will most likely obtain a slot since they will have one of the highest CEE scores within that next priority group. Applicants who are not assigned a slot are not enrolled. Importantly, a slot assignment is not an offer of admission, but automatic enrollment. Enrollees are expected to attend the University the following semester. The University bars students who do not attend from applying again in the following two admission cycles. In addition, the University’s rules make transferring majors once enrolled almost impossible.²⁰ The admission cycle ends when all slots are filled.

For the 2020-1 cycle, applicants in priority groups 1 and 2 took 74.5% of the slots in the Bogotá Campus and applicants in group 4 only took 225 slots or 7.4%.

A.5 Major-Specific Cutoffs

The admission cutoff for each major is determined by the score of the applicant who obtains the last available seat in that major. Cutoffs are therefore determined during the allocation process and are not known when applicants submit majors. The university does not provide real-time information on applicants’ distance from the current cutoff for their intended major or any other major.

¹⁹Most applicants (92.9%) from groups 1, 2 and 3 who are not enrolled in the first 24-hour period, but could log in to the next 24-period, choose not to do so.

²⁰Rules specified here: http://www.legal.unal.edu.co/rlunal/home/doc.jsp?d_i=66330#0.

After the end of each admission cycle, the university publishes the list of major-specific cutoffs online. Cutoffs from previous cycles are therefore publicly available before applicants submit majors in the current cycle. Applicants can compare their own score with previous cutoffs, but they do not know the number of applicants in their priority group, the majors those applicants will submit, or the realized cutoff for any major in the current cycle.